

Momentum-Space QCD Sum-Rule Setup for the $B_c(1P)$ Axial-Vector Mixing Angle

1 Purpose and conventions

We study the mixing between the two axial-vector $B_c(1P)$ basis states associated with the 3P_1 and 1P_1 structures. The current implementation follows the Aliev et al. mixing-angle method, but the OPE is evaluated in momentum space because both quark lines are heavy.

This note records the calculation implemented in `BcMixingMomentum.wl`. It is written so that every convention and intermediate object can be checked independently before using the numerical angle in a paper.

The quark content convention is

$$B_c^+ = \bar{b}c. \quad (1)$$

The two interpolating currents are taken as

$$J_\mu^A = \bar{b} \gamma_\mu \gamma_5 c, \quad (2)$$

$$J_\mu^B = i \bar{b} \sigma_{\mu\alpha} \frac{p^\alpha}{m_b + m_c} \gamma_5 c, \quad (3)$$

where

$$\sigma_{\mu\nu} = \frac{i}{2} [\gamma_\mu, \gamma_\nu]. \quad (4)$$

The explicit factor of i in J_μ^B is retained. This is important because `FeynCalc`'s `DiracSigma` already contains the conventional $i/2$ commutator.

The factor $1/(m_b + m_c)$ is also essential. Since $[\bar{Q}Q] = 3$ and $[p^\alpha] = 1$, the unnormalized tensor current $i\bar{b}\sigma_{\mu\alpha}p^\alpha\gamma_5c$ has mass dimension 4, while J_μ^A has mass dimension 3. The normalized current above makes both currents dimension 3.

If the opposite flavor convention is used, for example $B_c^- = \bar{c}b$, the sign of the quoted angle can change. The absolute convention must therefore be kept fixed when comparing to other papers.

2 Mass-dimension check

The quark field has dimension $3/2$, while p^α , m_b , and m_c have dimension 1. Therefore

$$[J_\mu^A] = 3, \quad [J_\mu^B] = 3 \quad (5)$$

only after the tensor current is normalized by $m_b + m_c$.

For currents with dimensions d_i and d_j ,

$$[\Pi_{\mu\nu}^{ij}] = d_i + d_j - 4, \quad (6)$$

because of the d^4x integration in the two-point function. Thus the normalized basis gives

$$[\Pi_1^{AA}] = [\Pi_1^{AB}] = [\Pi_1^{BB}] = 2. \quad (7)$$

The spectral densities have the same mass dimension as the invariant correlator,

$$[\rho^{AA}] = [\rho^{AB}] = [\rho^{BB}] = 2. \quad (8)$$

In the present code the Borel moment is evaluated as $\int ds e^{-s/M^2} \rho(s)$, so all three numerical moments carry one common extra factor of $[ds] = 2$. This common convention does not affect the mixing angle.

If the tensor current were left unnormalized, the dimensions would instead be

$$[\rho^{AA}], [\rho^{AB}], [\rho^{BB}] = 2, 3, 4, \quad (9)$$

and the matrix entries in the angle formula would not be dimensionally compatible.

3 Implementation map

The main code objects are:

- `Correlator[channel, order]`: $\Pi_1^{ij}(p^2)$ after FeynCalc Dirac algebra and spin projection.
- `SpectralDensity[channel, ‘pert’, s]`: $\rho_{\text{pert}}^{ij}(s)$.
- `NumericBorelPi[channel, order, M2, s0]`: $\Pi_{\text{order}}^{ij}(M^2, s_0)$.
- `NumericMixingAngleDegrees[M2, s0, ‘total’]`: $\theta(M^2, s_0)$ from $\Pi_{\text{pert}}^{ij} + \Pi_{G^2}^{ij}$.

The implemented OPE orders are

$$\text{pert}, \quad G^2, \quad \text{total} = \text{pert} + G^2. \quad (10)$$

The dimension-6 triple-gluon condensate $\langle g_s^3 G^3 \rangle$ is not included in the present file.

4 Correlators and spin-1 projection

Define

$$\Pi_{\mu\nu}^{ij}(p) = i \int d^4x e^{ip \cdot x} \langle 0 | T \{ J_\mu^i(x) J_\nu^{j\dagger}(0) \} | 0 \rangle, \quad i, j \in \{A, B\}. \quad (11)$$

The invariant amplitude used for the axial-vector spin-1 state is extracted with

$$P_{(1)}^{\mu\nu} = \frac{1}{3} \left(g^{\mu\nu} - \frac{p^\mu p^\nu}{p^2} \right), \quad (12)$$

so that

$$\Pi_1^{ij}(p^2) = P_{(1)}^{\mu\nu} \Pi_{\mu\nu}^{ij}(p). \quad (13)$$

The mixing angle is then obtained from

$$\tan 2\theta = -\frac{2\Pi_1^{AB}}{\Pi_1^{AA} - \Pi_1^{BB}}. \quad (14)$$

Numerically we use the quadrant-safe definition

$$\theta_{\text{raw}} = \frac{1}{2} \text{atan2}(-2\Pi_1^{AB}, \Pi_1^{AA} - \Pi_1^{BB}), \quad (15)$$

and quote the principal branch

$$\theta = \theta_{\text{raw}} - \frac{\pi}{2} \text{Round} \left(\frac{\theta_{\text{raw}}}{\pi/2} \right), \quad \theta \in [-45^\circ, 45^\circ]. \quad (16)$$

5 Momentum-space OPE

The perturbative heavy-quark propagator is

$$S_Q^{(0)}(k) = \frac{\not{k} + m_Q}{k^2 - m_Q^2}. \quad (17)$$

For the dimension-4 gluon condensate we use the background-field expansion

$$S_Q(k) = S_Q^{(0)}(k) + g_s G_A^{\alpha\beta} T^A S_{Q,\alpha\beta}^{(G)}(k) + S_Q^{(G^2)}(k), \quad (18)$$

where

$$S_{Q,\alpha\beta}^{(G)}(k) = -\frac{1}{4} \frac{\sigma_{\alpha\beta}(\not{k} + m_Q) + (\not{k} + m_Q)\sigma_{\alpha\beta}}{(k^2 - m_Q^2)^2}, \quad (19)$$

$$S_Q^{(G^2)}(k) = \frac{\langle g_s^2 G^2 \rangle}{12} m_Q \frac{k^2 + m_Q}{(k^2 - m_Q^2)^4}. \quad (20)$$

Equivalently,

$$g_s G_A^{\alpha\beta} T^A S_{Q,\alpha\beta}^{(G)}(k) = -\frac{g_s G_A^{\alpha\beta} T^A}{4} \frac{\sigma_{\alpha\beta}(\not{k} + m_Q) + (\not{k} + m_Q)\sigma_{\alpha\beta}}{(k^2 - m_Q^2)^2}. \quad (21)$$

The one-gluon fields are contracted with

$$\langle g_s^2 G_A^{\alpha\beta} G_B^{\rho\sigma} \rangle = \frac{\delta_{AB}}{96} \langle g_s^2 G^2 \rangle \left(g^{\alpha\rho} g^{\beta\sigma} - g^{\alpha\sigma} g^{\beta\rho} \right), \quad (22)$$

and

$$\text{Tr}(T^A T^B) = \frac{\delta^{AB}}{2}. \quad (23)$$

Thus the color factor for the $S^{(G)}S^{(G)}$ piece is

$$\frac{N_c^2 - 1}{192} \langle g_s^2 G^2 \rangle, \quad N_c = 3. \quad (24)$$

The three projected OPE contributions are generated from

$$\Pi_1^{AA} \sim P_{(1)}^{\mu\nu} \text{Tr} [\gamma_\mu \gamma_5 S_c(k) \gamma_\nu \gamma_5 S_b(k-p)], \quad (25)$$

$$\Pi_1^{AB} \sim P_{(1)}^{\mu\nu} \text{Tr} \left[\gamma_\mu \gamma_5 S_c(k) i\sigma_{\nu\alpha} \frac{p^\alpha}{m_b + m_c} \gamma_5 S_b(k-p) \right], \quad (26)$$

$$\Pi_1^{BB} \sim P_{(1)}^{\mu\nu} \text{Tr} \left[i\sigma_{\mu\alpha} \frac{p^\alpha}{m_b + m_c} \gamma_5 S_c(k) i\sigma_{\nu\beta} \frac{p^\beta}{m_b + m_c} \gamma_5 S_b(k-p) \right]. \quad (27)$$

6 Perturbative spectral densities

For the perturbative loop, impose the two-particle cut

$$k^2 = m_c^2, \quad (k-p)^2 = m_b^2, \quad (28)$$

which gives

$$k \cdot p = \frac{s + m_c^2 - m_b^2}{2}, \quad s = p^2. \quad (29)$$

Define

$$\lambda(s, m_b, m_c) = s^2 + m_b^4 + m_c^4 - 2sm_b^2 - 2sm_c^2 - 2m_b^2 m_c^2. \quad (30)$$

The perturbative spectral densities are implemented as

$$\rho_{\text{pert}}^{ij}(s) = \frac{N_c}{16\pi^2} \frac{\sqrt{\lambda(s, m_b, m_c)}}{s} \frac{N^{ij}(s)}{(m_b + m_c)^{n_B^{ij}}}, \quad (31)$$

where $n_B^{AA} = 0$, $n_B^{AB} = n_B^{BA} = 1$, and $n_B^{BB} = 2$ counts how many tensor currents occur in the channel. The unnormalized numerator functions are

$$N^{AA}(s) = -\frac{2}{s} [(m_b + m_c)^2 - s] [(m_b - m_c)^2 + 2s], \quad (32)$$

$$N^{AB}(s) = 6(m_b - m_c) [(m_b + m_c)^2 - s], \quad (33)$$

$$N^{BB}(s) = 2 [-2(m_b^2 - m_c^2)^2 + (m_b^2 - 6m_b m_c + m_c^2)s + s^2]. \quad (34)$$

Equivalently, after setting $N_c = 3$,

$$\rho_{\text{pert}}^{AA}(s) = -\frac{3}{8\pi^2 s^2} [(m_b + m_c)^2 - s] [(m_b - m_c)^2 + 2s] \sqrt{\lambda(s, m_b, m_c)}, \quad (35)$$

$$\rho_{\text{pert}}^{AB}(s) = \frac{9}{8\pi^2 s(m_b + m_c)} (m_b - m_c) [(m_b + m_c)^2 - s] \sqrt{\lambda(s, m_b, m_c)}, \quad (36)$$

$$\rho_{\text{pert}}^{BB}(s) = \frac{3}{8\pi^2 s(m_b + m_c)^2} [-2(m_b^2 - m_c^2)^2 + (m_b^2 - 6m_b m_c + m_c^2)s + s^2] \sqrt{\lambda(s, m_b, m_c)}. \quad (37)$$

The perturbative Borel moments are

$$\Pi_{\text{pert}}^{ij}(M^2, s_0) = \int_{(m_b+m_c)^2}^{s_0} ds e^{-s/M^2} \rho_{\text{pert}}^{ij}(s). \quad (38)$$

With this convention all three perturbative spectral densities have mass dimension 2. Without the factors of $(m_b + m_c)^{-n_B^{ij}}$, their dimensions would be 2, 3, and 4 for AA , AB , and BB , respectively.

7 Dimension-4 gluon condensate

The implemented condensate contribution is

$$\Pi_{G^2}^{ij} = \Pi^{ij}[S_c^{(G^2)} S_b^{(0)}] + \Pi^{ij}[S_c^{(0)} S_b^{(G^2)}] + \Pi^{ij}[S_c^{(G)} S_b^{(G)}]. \quad (39)$$

After Feynman parametrization the terms have the schematic form

$$\mathcal{A}_{G^2}^{ij}(x_1, x_2, s) = \sum_r C_r^{ij}(x_1, x_2, s) [m_c^2 x_1^2 + (m_b^2 + m_c^2 - s)x_1 x_2 + m_b^2 x_2^2]^{-n_r}. \quad (40)$$

The code then fixes the simplex by

$$x_1 = x, \quad x_2 = 1 - x. \quad (41)$$

This gives

$$\Delta(x, s) = m_c^2 x + m_b^2 (1 - x) - s x (1 - x). \quad (42)$$

The G^2 terms are not treated as ordinary smooth spectral densities. We instead use a direct Borel transform. Set $s = -Q^2$ and write

$$\Delta(x, -Q^2) = x(1 - x) [Q^2 + \bar{s}(x)], \quad (43)$$

where

$$\bar{s}(x) = \frac{m_c^2 x + m_b^2 (1 - x)}{x(1 - x)}. \quad (44)$$

The Feynman-parameter amplitude is decomposed as

$$-i \mathcal{A}_{G^2}^{ij}(x, -Q^2) = \sum_{n \geq 1} \frac{a_n^{ij}(x)}{[Q^2 + \bar{s}(x)]^n} + \text{polynomial subtraction terms.} \quad (45)$$

The polynomial subtraction terms vanish after the Borel transform. The working phase convention is the explicit factor $-i$ shown above. This is a bookkeeping convention for removing the loop-integration prefactor returned by FeynCalc's Feynman-parameter amplitude. This is the convention that should be independently checked.

We use

$$\mathcal{B}_{M^2} \left[\frac{1}{(Q^2 + \bar{s})^n} \right] = \frac{1}{(n-1)! (M^2)^{n-1}} e^{-\bar{s}/M^2}. \quad (46)$$

Thus

$$\Pi_{G^2}^{ij}(M^2, s_0) = \int_{x_-}^{x_+} dx e^{-\bar{s}(x)/M^2} \sum_{n \geq 1} \frac{a_n^{ij}(x)}{(n-1)! (M^2)^{n-1}}. \quad (47)$$

The continuum restriction is

$$\bar{s}(x) \leq s_0, \quad (48)$$

which gives

$$x_{\pm} = \frac{s_0 + m_b^2 - m_c^2 \pm \sqrt{\lambda(s_0, m_b, m_c)}}{2s_0}. \quad (49)$$

8 Total Borel moments and numerical test point

The total moments used in the current numerical result are

$$\Pi^{ij}(M^2, s_0) = \Pi_{\text{pert}}^{ij}(M^2, s_0) + \Pi_{G^2}^{ij}(M^2, s_0). \quad (50)$$

The current default numerical inputs are

$$m_b = 4.18 \text{ GeV}, \quad (51)$$

$$m_c = 1.27 \text{ GeV}, \quad (52)$$

$$\langle g_s^2 G^2 \rangle = 4\pi^2 (0.012) \text{ GeV}^4, \quad (53)$$

$$M^2 = 10 \text{ GeV}^2, \quad (54)$$

$$s_0 = 55 \text{ GeV}^2. \quad (55)$$

For this point,

$$x_- = 0.3322728664, \quad x_+ = 0.9560816791. \quad (56)$$

The perturbative moments are

$$\Pi_{\text{pert}}^{AA} = 0.1441716826, \quad (57)$$

$$\Pi_{\text{pert}}^{AB} = -0.1051577192, \quad (58)$$

$$\Pi_{\text{pert}}^{BB} = 0.1348134513. \quad (59)$$

The gluon-condensate moments are

$$\Pi_{G^2}^{AA} = -0.0003658277, \quad (60)$$

$$\Pi_{G^2}^{AB} = 0.0002309420, \quad (61)$$

$$\Pi_{G^2}^{BB} = -0.0001888869. \quad (62)$$

Thus

$$\Pi_{\text{total}}^{AA} = 0.1438058549, \quad (63)$$

$$\Pi_{\text{total}}^{AB} = -0.1049267772, \quad (64)$$

$$\Pi_{\text{total}}^{BB} = 0.1346245644. \quad (65)$$

The corresponding angles are

$$\theta_{\text{pert}} = 43.726118943^\circ, \quad (66)$$

$$\theta_{\text{pert}+G^2} = 43.747426660^\circ. \quad (67)$$

At this point the relative condensate corrections are

$$\frac{\Pi_{G^2}^{AA}}{\Pi_{\text{pert}}^{AA}} = -0.002537, \quad (68)$$

$$\frac{\Pi_{G^2}^{AB}}{\Pi_{\text{pert}}^{AB}} = -0.002196, \quad (69)$$

$$\frac{\Pi_{G^2}^{BB}}{\Pi_{\text{pert}}^{BB}} = -0.001401. \quad (70)$$

9 How to reproduce the numerical check

In Mathematica, load the implementation with

```
SetDirectory["/Users/sbilmis/Bc_mixing"];
Get["BcMixingMomentum.wl"];
```

Then evaluate

```
NumericOPESummary[10, 55]
NumericMixingAngleDegrees[10, 55, "pert"]
NumericMixingAngleDegrees[10, 55, "total"]
MixingMassDimensionReport[]
CheckMixingMassDimensions[]
PerturbativeSpectralDensityDimensionReport[]
```

For window studies one can scan

```
MixingAngleTable[{8, 12, 1}, {50, 60, 5}, "total"]
OPEConvergenceDataset[{8, 12, 1}, {50, 60, 5}]
```

10 Checklist for independent verification

A theorist can check the calculation in the following order:

1. Confirm the current convention, especially the explicit factor of i in J_μ^B , the normalization $p^\alpha/(m_b + m_c)$, and the definition of $\sigma_{\mu\nu}$.
2. Reproduce the three projected perturbative traces and verify the numerator functions $N^{AA}(s)$, $N^{AB}(s)$, and $N^{BB}(s)$.
3. Check the two-body phase-space normalization $N_c\sqrt{\lambda}/(16\pi^2s)$ used in $\rho_{\text{pert}}^{ij}(s)$.

4. Verify the background-field propagator terms and the color factor in the $S^{(G)}S^{(G)}$ contribution.
5. Independently derive the direct Borel formula for the G^2 terms, including the $-i$ phase convention.
6. Check that $\Pi_{G^2}/\Pi_{\text{pert}}$ remains small in the accepted Borel window.
7. Determine the final working window using OPE convergence, Borel stability, and pole-dominance criteria.